
paper_id: PAPER_1003
title: "Spectral Ladder Merger Coupling — 26-State Energy Hierarchy"
session: 219
date: 2026-04-13
author: "Daniel T. Murphy"
status: production
cvw: "v2.0.0"
tags: [spectral, ladder, merger, coupling, 26D, energy, hierarchy]
crosslinks: [PAPER_958, PAPER_999, PAPER_1001]
calibration: {E0: " $\hbar\omega_{\text{SCm}}$ ", states: 26, S26_3rd: 0.095}
sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_1003: Spectral Ladder Merger Coupling

Abstract

We apply the 26-state spectral ladder $E_n = \hbar\omega_{\text{SCm}} \cdot (2\pi)^{\{n/3\}} \cdot S_{26}^{(3)}$ to merger dynamics. The ladder encodes frequency-dependent energy injection into the merger waveform across 26 independent dimensional channels.

1. Spectral Energy

$$E_{\text{ladder}} = \sum_{\{n=1\}}^{\{26\}} E_n = \sum \hbar\omega_{\text{SCm}} \cdot (2\pi)^{\{n/3\}} \cdot S_{26}^{(3)}$$

Result: $E_{\text{ladder}} = 1.42 \times 10^{-15} \text{ J}$.

2. Jet Modulation

$M_{\text{jet}}(\Gamma_0) = 1 + A_{\text{jet}} \cdot \exp(0)$ gives resonance modulation factor of $2.5\times$ at $A_{\text{jet}} = 1.5$.

3. Implementation

File: `fubi_agn_ns_mergers.py`, class `SpectralLadderMergerCalc`. CP4 class #587.